PROTAMINE SULFATE

Slow IV — hypotension
Fish allergy caution

CHEMICAL NAME Mixture of low-molecular-weight, arginine-rich basic proteins from fish (salmon) sperm, as sulfate
CLASS Heparin antagonist (antidote)
TRADE NAMES Protamine sulfate injection · Prosulf

THERAPEUTIC USES / INDICATIONS

- Reversal of unfractionated heparin (overdose, after cardiopulmonary bypass, bleeding).
- Partial reversal of LMWH (enoxaparin); ~60% of anti-Xa activity.

ADMINISTRATION

- **IV slowly over 10 min** — max 5 mg/min; never rapid.
- Resuscitation equipment available.
- May dilute in saline or dextrose.

DOSAGES

- **UFH:** 1 mg per 100 units heparin given in the last 15 min (less if longer ago — heparin t½ ~1 h); max 50 mg per dose.
- **After 30–60 min:** 0.5–0.75 mg per 100 units; >2 h: 0.25–0.375 mg per 100 units.
- **Enoxaparin (within 8 h):** 1 mg per 1 mg enoxaparin; second dose 0.5 mg per 1 mg if bleeding continues.

PHARMACOKINETICS

- **Onset:** 5 min. **Duration:** ~2 h (shorter than some heparin — rebound possible).
- Fate not fully known (partly degraded by fibrinolysin).

PHARMACODYNAMICS

- Strongly basic protein binds acidic heparin → stable inactive salt.
- In excess, protamine has its own weak anticoagulant effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flushing, feeling of warmth.
- Nausea, vomiting.
- Bradycardia.

ADVERSE EFFECTS (SERIOUS)

- Severe hypotension, anaphylaxis, pulmonary hypertension, bronchospasm (rapid injection, fish allergy, previous protamine/NPH insulin exposure, vasectomy).
- Rebound heparin effect (bleeding hours later).
- Excess dose → anticoagulation.

CONTRAINDICATIONS & CAUTIONS

- Previous severe reaction to protamine.
- **Caution:** fish allergy, previous exposure to protamine-containing insulin, infertile/vasectomised men.

KEY INTERACTIONS

- Protamine-containing insulins (NPH) may sensitise to reactions.

NURSING MANAGEMENT

- Calculate dose from heparin given and timing; double-check.
- Infuse slowly; monitor BP, HR, SpO_2 continuously.
- Check aPTT/ACT 5–15 min after and again 2–8 h later (rebound).
- Monitor for bleeding.
- Have adrenaline available.

PATIENT EDUCATION

- Report flushing, breathlessness or dizziness during injection.

KEY POINTS

- Heparin antidote; 1 mg per 100 units (max 50 mg).
- Slow IV — hypotension, anaphylaxis.
- Only partial reversal of LMWH.
- Recheck aPTT for rebound.